

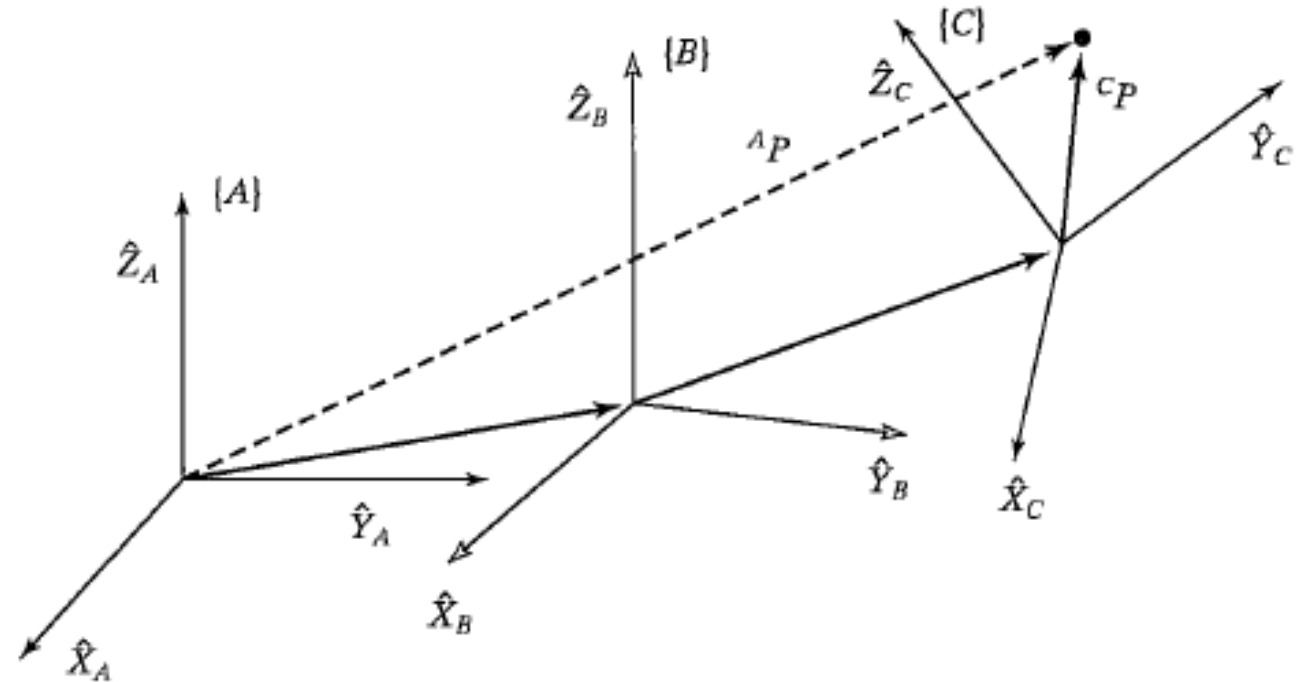


RAI 811: ROBOT MECHANICS & CONTROL

Compound Transformations



- Frame $\{C\}$ is known relative to frame $\{B\}$.
- Frame $\{B\}$ is known relative to frame $\{A\}$.
- We can transform point P from frame $\{C\}$ to frame $\{A\}$.



Compound Transformations



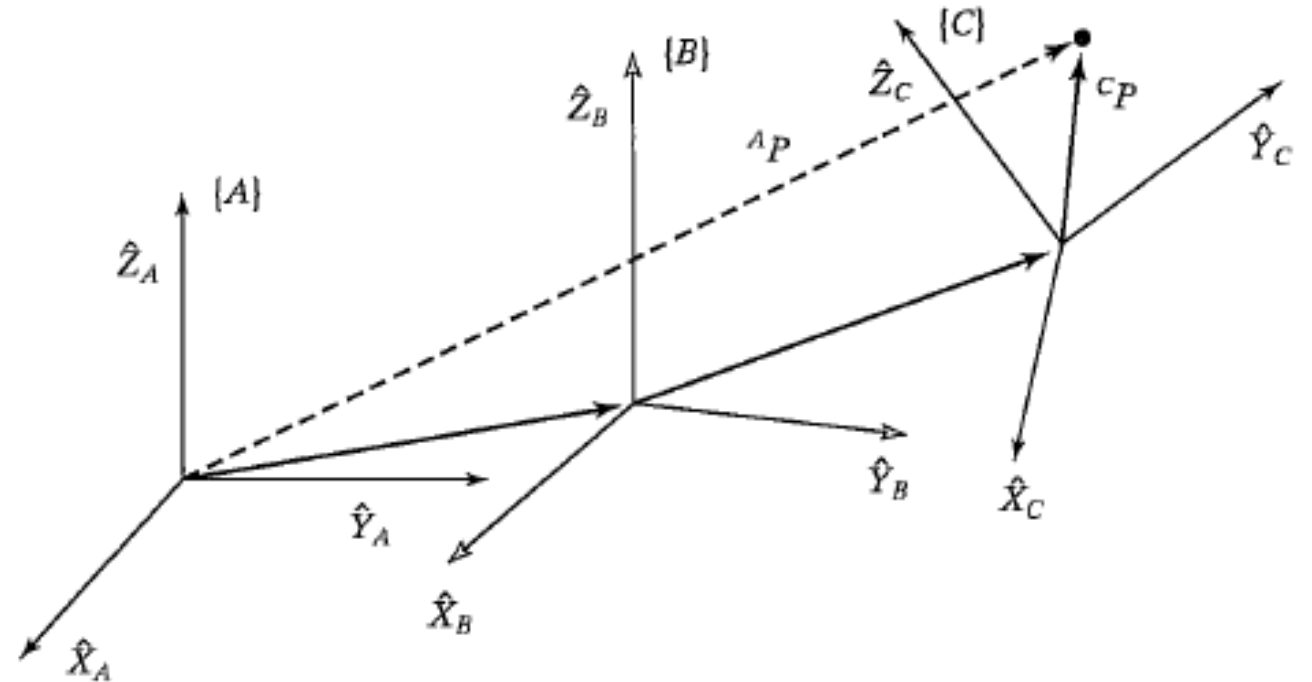
$${}^B P = {}^B T {}^C P$$

$$\Rightarrow {}^A P = {}^A T {}^C P$$

$${}^A P = {}^A T {}^B T {}^C P$$

From here define

$${}^A T {}^C T = {}^A T {}^B T {}^C T$$



Inverting a Transform

- Suppose that a frame {B} is known relative to frame {A}.
It means that we know ${}^A_B T$.
- We want to calculate its inverse.
We want to know frame {A} relative to frame {B}.
It means that we want to calculate ${}^B_A T$.

Inverting a Transform

To find ${}^B_A T$, we must compute ${}^B_A R$ and ${}^B P_{AORG}$ from ${}^A_R R$ and ${}^A P_{BORG}$.

recall from our discussion of rotation matrices that

$${}^B_A R = {}^A_B R^T.$$

Next, we change the description of ${}^A P_{BORG}$ into $\{B\}$ by using (2.13):

$${}^A P = {}^A_B R {}^B P. \quad (2.13)$$

$${}^B ({}^A P_{BORG}) = {}^B_A R {}^A P_{BORG} + {}^B P_{AORG}. \quad (2.43)$$

Inverting a Transform

The left-hand side of (2.43) must be zero, so we have

$${}^B P_{AORG} = -{}^B_A R {}^A P_{BORG} = -{}^A_B R^T {}^A P_{BORG}. \quad (2.44)$$

Using (2.42) and (2.44), we can write the form of ${}^B_A T$ as

$${}^B_A T = \left[\begin{array}{ccc|c} {}^A_B R^T & -{}^A_B R^T {}^A P_{BORG} & & \\ \hline 0 & 0 & 0 & 1 \end{array} \right]. \quad (2.45)$$

Note that, with our notation,

$${}^B_A T = {}^A_B T^{-1}.$$

Example 2.5

Figure 2.13 shows a frame $\{B\}$ that is rotated relative to frame $\{A\}$ about $\hat{Z}$ by 30 degrees and translated four units in $\hat{X}_A$ and three units in $\hat{Y}_A$. Thus, we have a description of ${}^A_B T$. Find ${}^B_A T$.

The frame defining $\{B\}$ is

$${}^A_B T = \begin{bmatrix} 0.866 & -0.500 & 0.000 & 4.0 \\ 0.500 & 0.866 & 0.000 & 3.0 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.46)$$

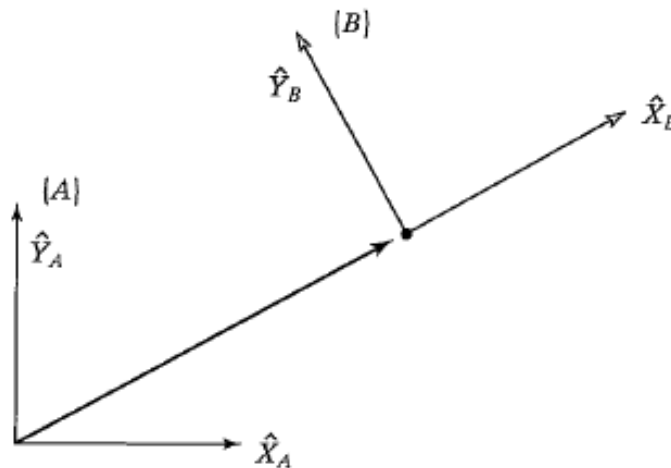


FIGURE 2.13: $\{B\}$ relative to $\{A\}$.

Example 2.5



Using (2.45), we compute

$${}^B_A T = \begin{bmatrix} 0.866 & 0.500 & 0.000 & -4.964 \\ -0.500 & 0.866 & 0.000 & -0.598 \\ 0.000 & 0.000 & 1.000 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

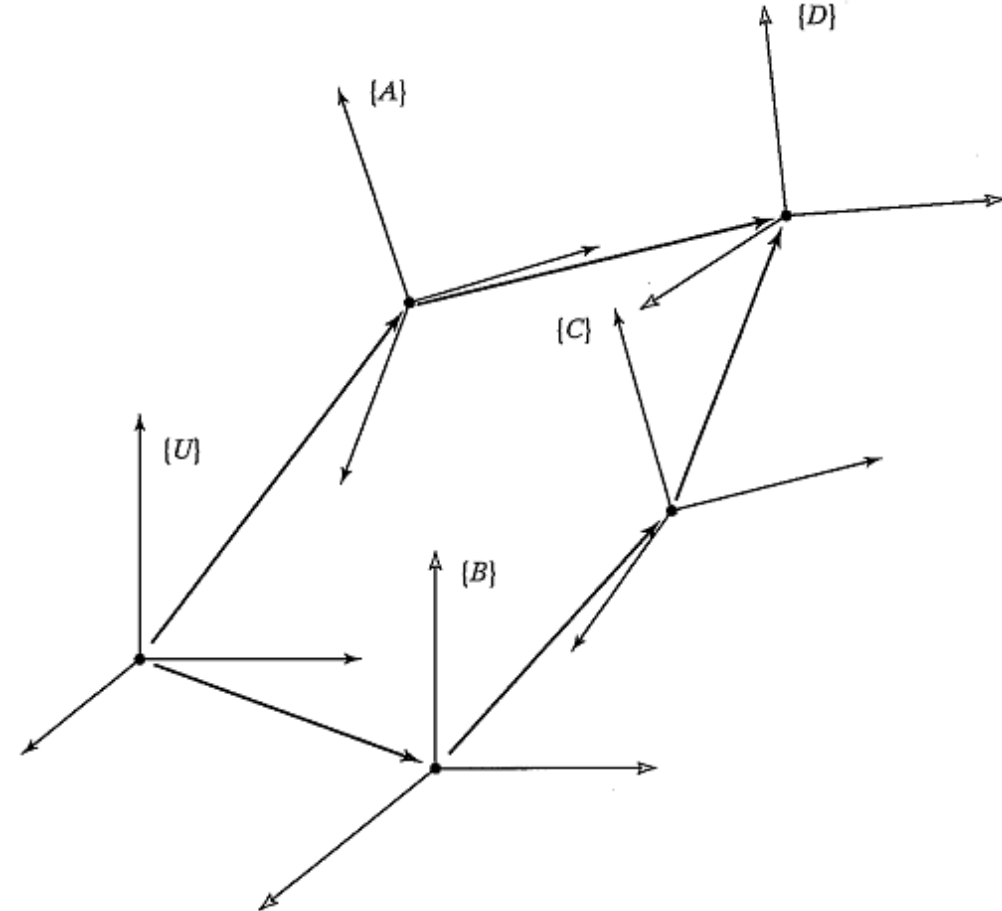
Transform Equations

$${}^U T_D = {}^U T_A {}^A T_D$$

$${}^U T_D = {}^U T_B {}^B T_C {}^C T_D$$

$${}^U T_A {}^A T_D = {}^U T_B {}^B T_C {}^C T_D$$

$${}^B T_C = {}^U T_B^{-1} {}^U T_A {}^A T_D {}^C T_D^{-1}$$

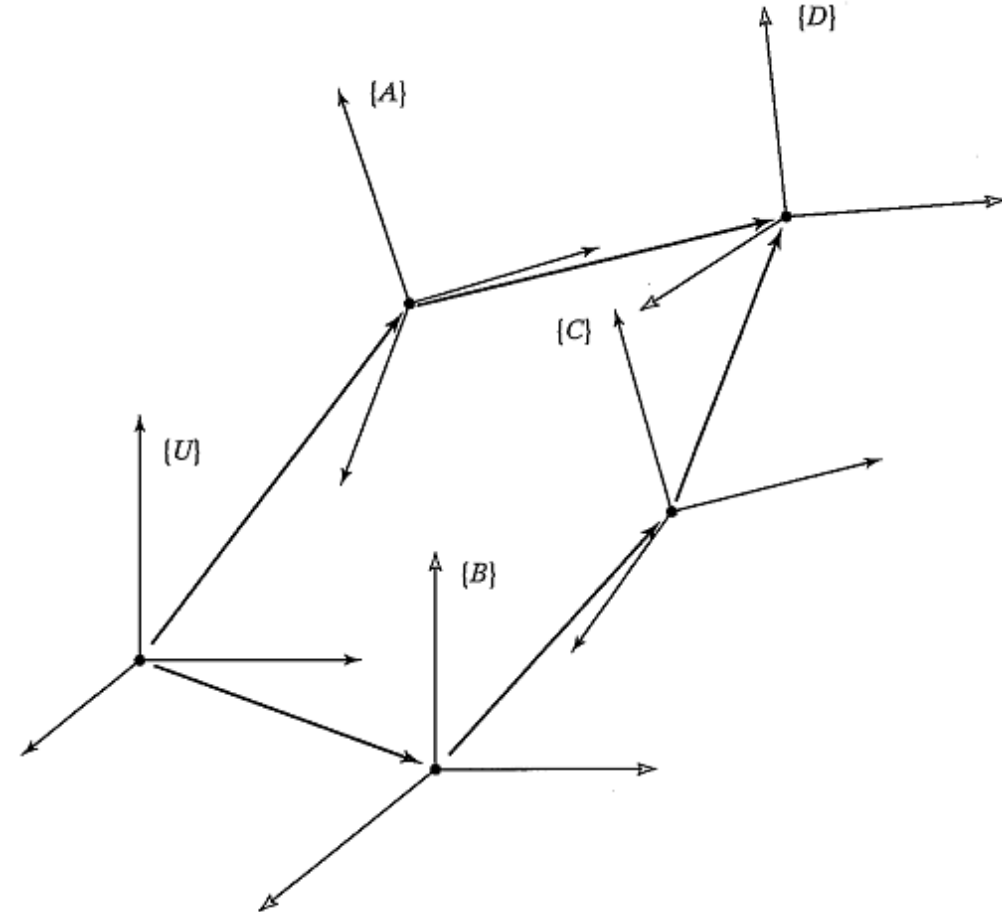


Transform Equations

$${}^U_T = {}^U_A {}^A_D {}^D_C {}^C_T$$

$${}^U_T = {}^U_B {}^B_C {}^C_T.$$

$${}^U_A = {}^U_B {}^B_C {}^C_D {}^D_A {}^D_T.$$



Transform Equations



$${}^G T_T = {}^B T_T^{-1} {}^B T_S {}^S T_G.$$

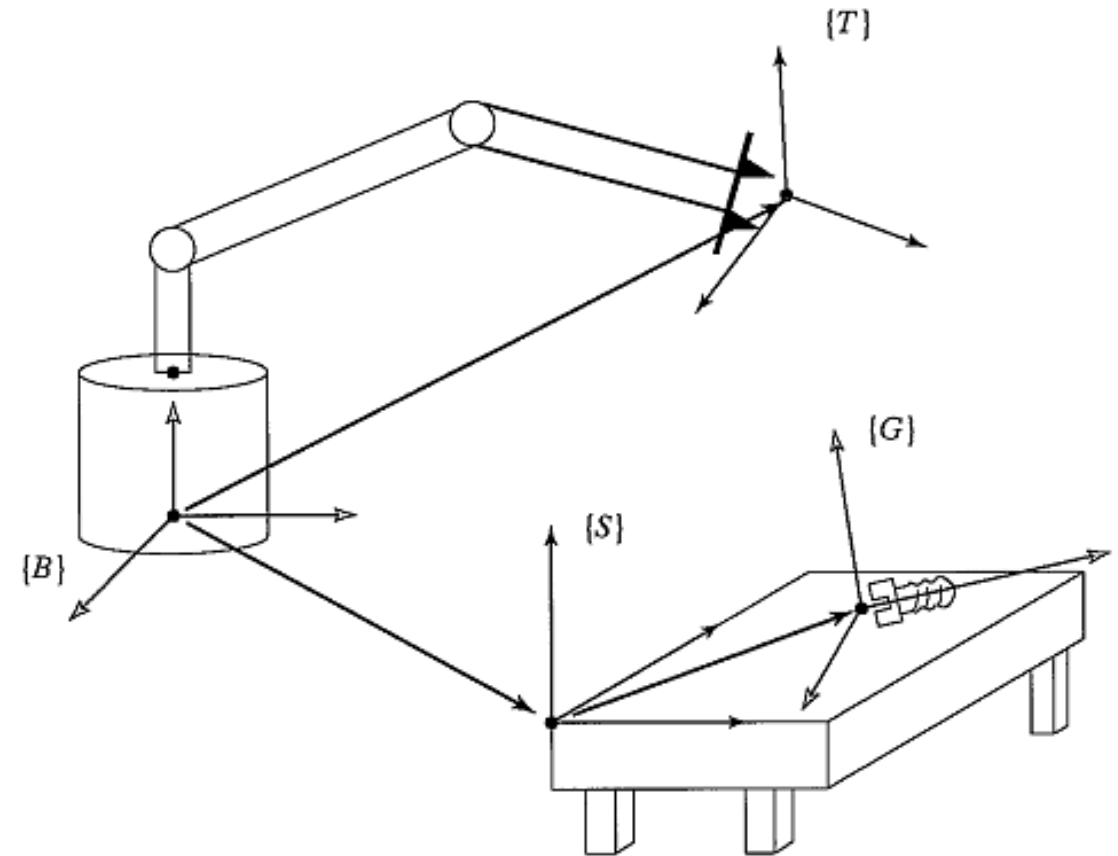


FIGURE 2.16: Manipulator reaching for a bolt.

Example 2.7

- Rotation is not commutative.

Consider two rotations, one about $\hat{Z}$ by 30 degrees and one about $\hat{X}$ by 30 degrees:

$$R_z(30) = \begin{bmatrix} 0.866 & -0.500 & 0.000 \\ 0.500 & 0.866 & 0.000 \\ 0.000 & 0.000 & 1.000 \end{bmatrix} \quad (2.60)$$

$$R_x(30) = \begin{bmatrix} 1.000 & 0.000 & 0.000 \\ 0.000 & 0.866 & -0.500 \\ 0.000 & 0.500 & 0.866 \end{bmatrix} \quad (2.61)$$

$$\begin{aligned} R_z(30)R_x(30) &= \begin{bmatrix} 0.87 & -0.43 & 0.25 \\ 0.50 & 0.75 & -0.43 \\ 0.00 & 0.50 & 0.87 \end{bmatrix} \\ \neq R_x(30)R_z(30) &= \begin{bmatrix} 0.87 & -0.50 & 0.00 \\ 0.43 & 0.75 & -0.50 \\ 0.25 & 0.43 & 0.87 \end{bmatrix} \end{aligned} \quad (2.62)$$

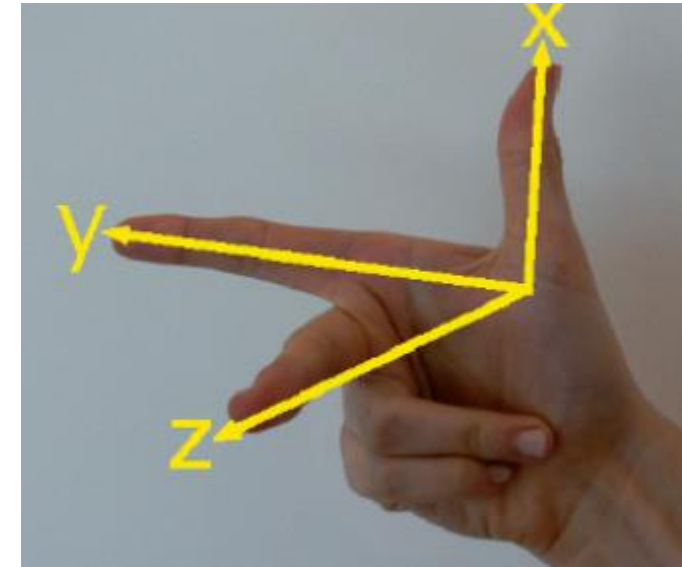
Working in 3D

A point **P** is represented by its x -, y - and z -coordinates (x, y, z) or as a bound vector

$$\mathbf{p} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

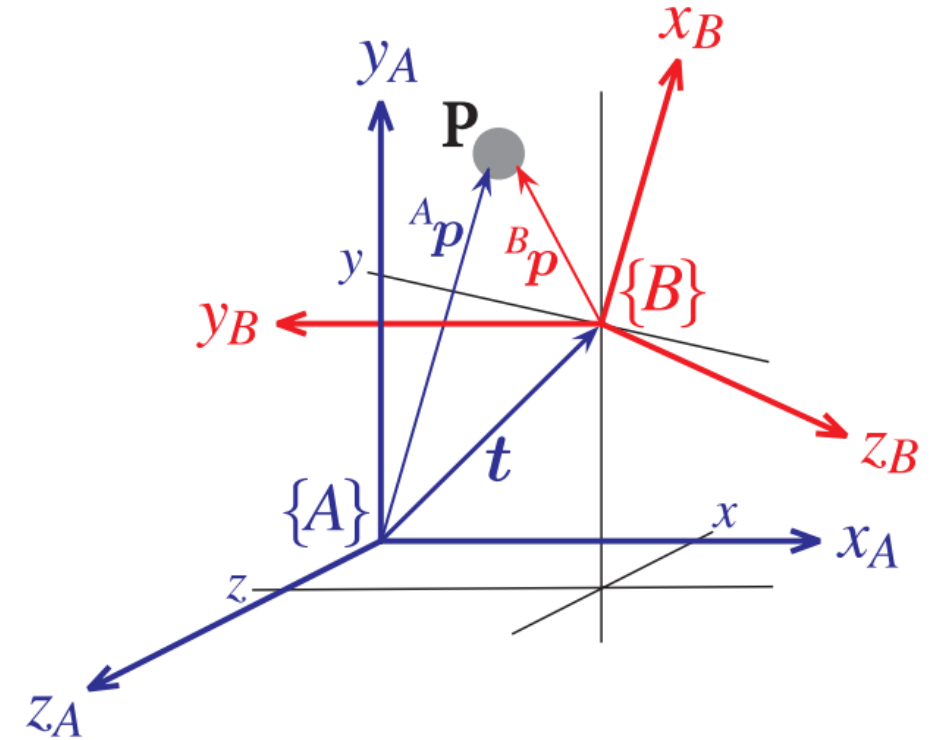
Unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$

$$\hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \quad \hat{\mathbf{x}} = \hat{\mathbf{y}} \times \hat{\mathbf{z}}; \quad \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$$



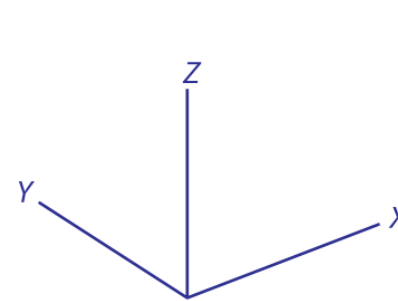
Working in 3D

- Description of frame $\{B\}$ w.r.t. frame $\{A\}$?
- We consider an arbitrary point P w.r.t. to each of the coordinate frames and determine the relationship between ${}^A P$ and ${}^B P$ to obtain translation and orientation.
- Translation is easy whereas orientation for 3D is difficult

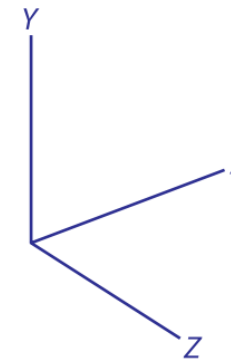


Orientation in 3D

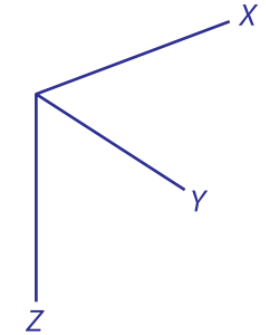
- *Euler's rotation theorem* states that any rotation can be considered as a sequence of rotations about different coordinate axes.
- We start by considering rotation about a single coordinate axis.



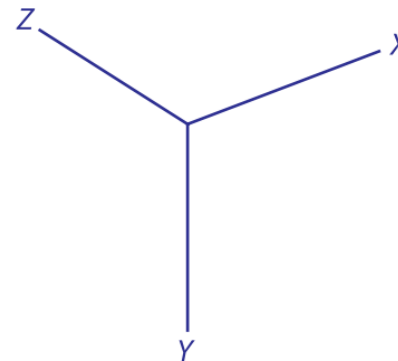
a Original



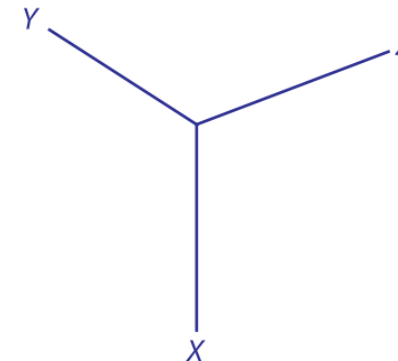
b $\frac{\pi}{2}$ about x-axis



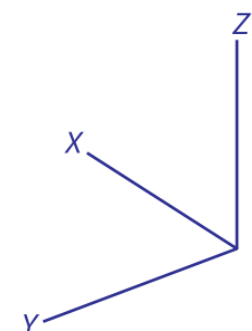
c π about x-axis



d $-\frac{\pi}{2}$ about x-axis

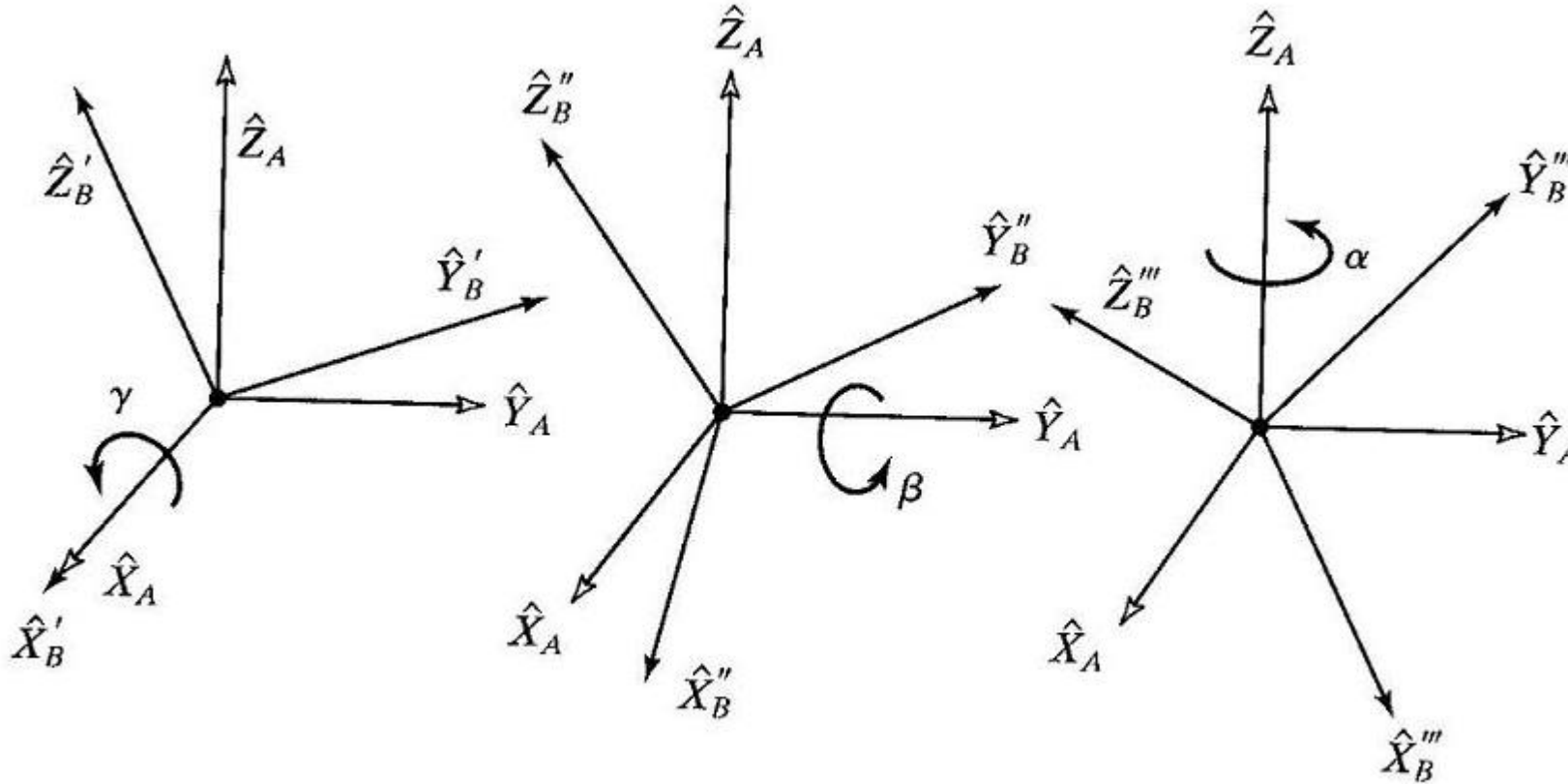


e $\frac{\pi}{2}$ about y-axis



f $\frac{\pi}{2}$ about z-axis

XYZ Fixed Angles

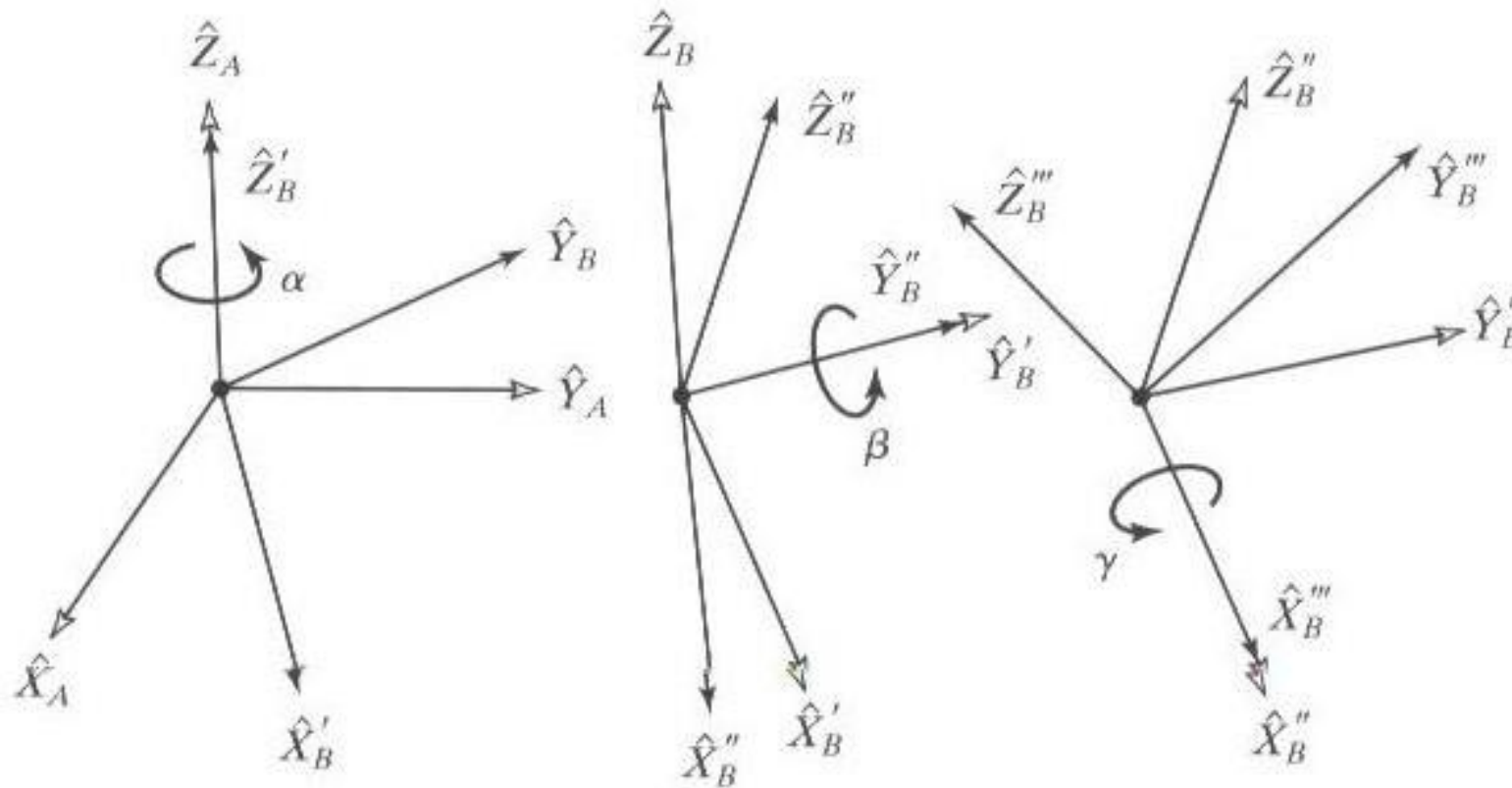


XYZ Fixed Angles

$$\begin{aligned} {}^A_B R_{XYZ}(\gamma, \beta, \alpha) &= R_Z(\alpha) R_Y(\beta) R_X(\gamma) \\ &= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}, \quad (2.63) \end{aligned}$$

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}. \quad (2.64)$$

ZYX Euler Angles



ZYX Euler Angles

$$\begin{aligned}
 {}^A_B R_{Z'Y'X'} &= R_Z(\alpha) R_Y(\beta) R_X(\gamma) \\
 &= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}, \quad (2.70)
 \end{aligned}$$

$${}^A_B R_{Z'Y'X'}(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}. \quad (2.71)$$

Note that the result is exactly the same as that obtained for the same three rotations taken in the opposite order about fixed axes! This somewhat nonintuitive result holds in general: three rotations taken about fixed axes yield the same final orientation as the same three rotations taken in opposite order about the axes of the moving frame.

Exercise 2.27



Referring to Fig. 2.25, give the value of ${}^A_B T$.

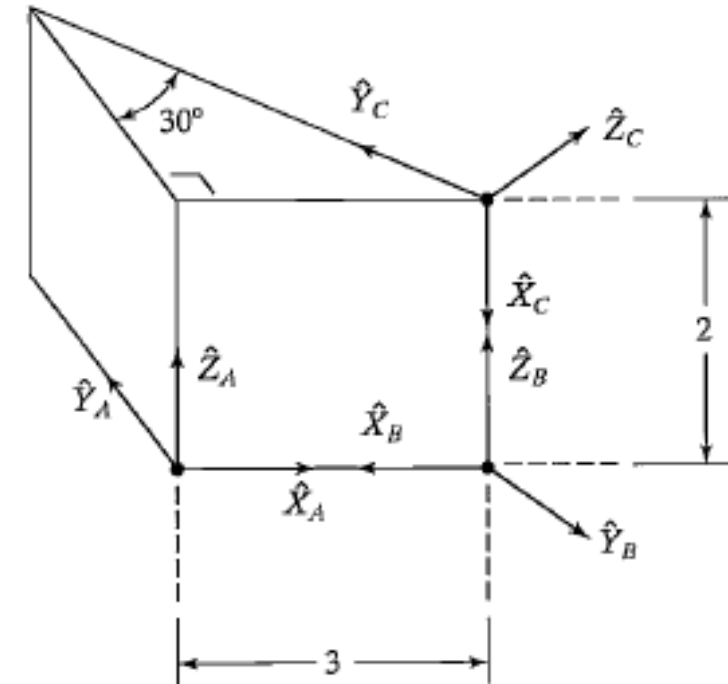


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.27

$${}^{Borg}P = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} \quad {}^A_B R = R_z(180^\circ)$$

$${}^A_B R = \begin{bmatrix} \cos(180^\circ) & -\sin(180^\circ) & 0 \\ \sin(180^\circ) & \cos(180^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^A_B R = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

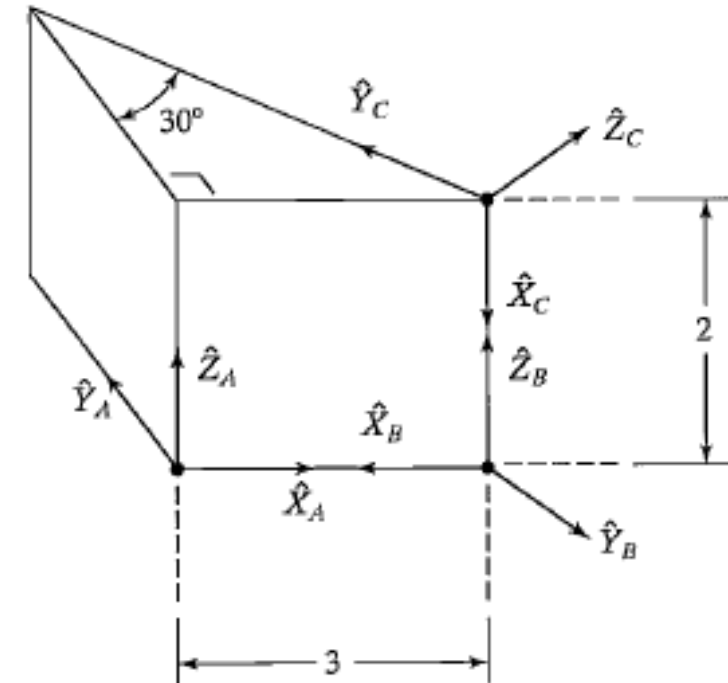


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.27

$${}^A_B T = \begin{bmatrix} {}^A_B R & \text{Borg}^A_P \\ 0 & 1 \end{bmatrix}$$

$${}^A_B T = \begin{bmatrix} -1 & +0 & +0 & 3 \\ +0 & -1 & +0 & 0 \\ +0 & +0 & +1 & 0 \\ +0 & +0 & +0 & 1 \end{bmatrix}$$

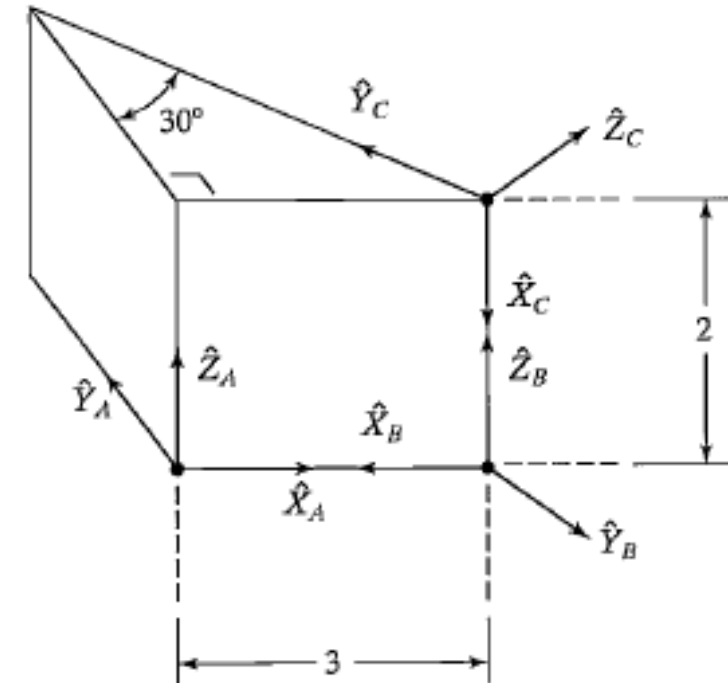


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.29



Referring to Fig. 2.25, give the value of ${}^B_C T$

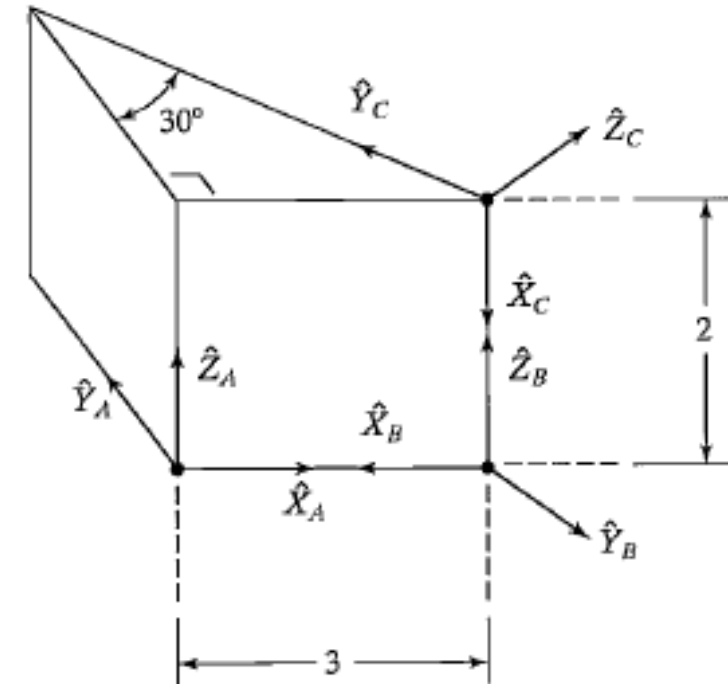


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.29

$${}^{Corg}{}^B P = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$${}^B{}_C R = R_y(90^\circ) R_x(150^\circ)$$

$$R_y(90^\circ) = \begin{bmatrix} \cos(90^\circ) & 0 & \sin(90^\circ) \\ 0 & 1 & 0 \\ -\sin(90^\circ) & 0 & \cos(90^\circ) \end{bmatrix}$$

$$R_y(90^\circ) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

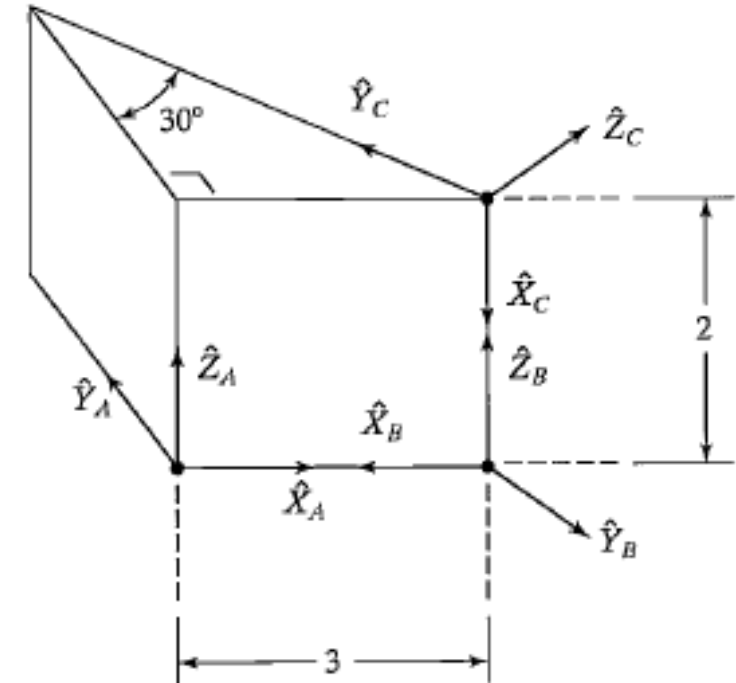


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.29

$$R_x(150^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(150^\circ) & -\sin(150^\circ) \\ 0 & \sin(150^\circ) & \cos(150^\circ) \end{bmatrix}$$

$$R_x(150^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.866 & -0.5 \\ 0 & 0.5 & -0.866 \end{bmatrix}$$

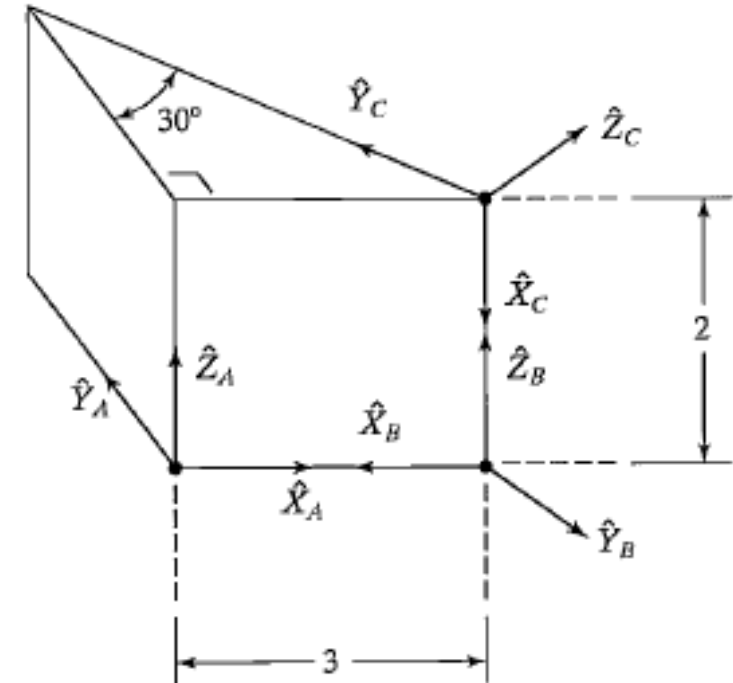


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.29

$${}^B_C R = R_y(90^\circ)R_x(150^\circ)$$

$${}^B_C R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.866 & -0.5 \\ 0 & 0.5 & -0.866 \end{bmatrix}$$

$${}^B_C R = \begin{bmatrix} 0 & 0.5 & -0.866 \\ 0 & -0.866 & -0.5 \\ -1 & 0 & 0 \end{bmatrix}$$

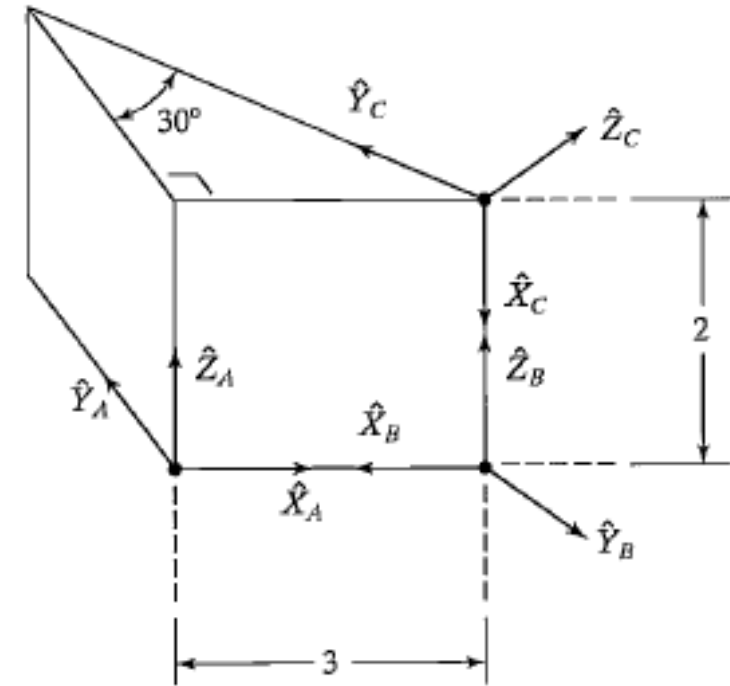


FIGURE 2.25: Frames at the corners of a wedge.

Exercise 2.29

$${}^B_C T = \begin{bmatrix} {}^B_C R & \text{Cor}g^B_P \\ 0 & 1 \end{bmatrix}$$

$${}^B_C T = \begin{bmatrix} +0.000 & +0.500 & -0.866 & 0 \\ +0.000 & -0.866 & -0.500 & 0 \\ -1.000 & +0.000 & +0.000 & 2 \\ +0.000 & +0.000 & +0.000 & 1 \end{bmatrix}$$

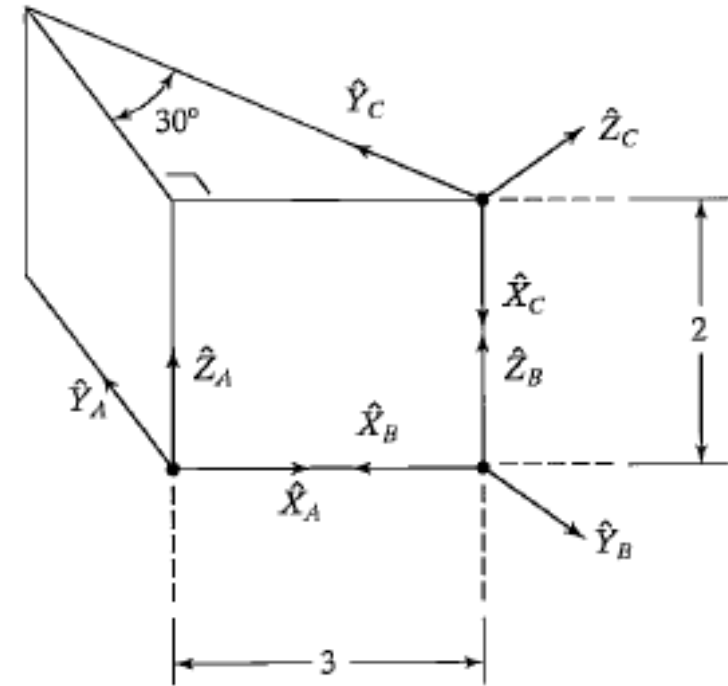


FIGURE 2.25: Frames at the corners of a wedge.

Any Questions?

